Expt. 6: Antenna Characterization (Lab Report)

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Contents

1 Aim **3**

2 Apparatus Required **3**

3 Theory **3**
3.1 Basic Antenna Concepts . . . 3
3.2 Antenna Radiation Pattern . . . 4
3.3 Reflection Coefficient, VSWR, and Return Loss . . . 4
3.4 Effect of Distance on Received Power . . . 5
3.5 Co-Polarization and Cross-Polarization . . . 5
3.6 Half-Power Beamwidth . . . 6
3.7 Antenna Directivity . . . 6

4 Experimental Setup **8**
4.1 VSWR and Operating-Frequency Measurement . . . 8
4.2 Received-Power Measurement . . . 8
4.3 Beam-Pattern Measurement . . . 8

5 Experimental Procedure **9**
5.1 Part A: VSWR and Antenna Operating Frequency . . . 9
5.2 Part B: Effect of Distance on Received Power . . . 9
5.3 Part C: Beam Pattern in Co-Polarization . . . 10
5.4 Part D: Beam Pattern in Cross-Polarization . . . 10
5.5 Part E: Half-Power Beamwidth and Directivity . . . 11

6 Observations **12**
6.1 Observation Table: VSWR Measurement . . . 12
6.2 Recorded Antenna-Length Calculation . . . 12
6.3 Observation Table: Received Power versus Distance . . . 12
6.4 Observation Table: Co-Polarization . . . 14
6.5 Observation Table: Cross-Polarization . . . 14

7 Results **16**
7.1 Antenna Operating Frequency 16
7.2 Effect of Distance on Received Power 16
7.3 Co-Polarization Beam Pattern 17
7.4 Cross-Polarization Beam Pattern 17

8 Key Takeaways **18**

1 Aim

(a) To determine the antenna operating frequency and return loss from VSWR measurements.

(b) To measure the effect of distance on the power received by the antenna.

(c) To measure the far-field power pattern of the given antenna with respect to angle in both co-polarization and cross-polarization configurations.

(d) To determine the half-power beamwidth and estimate the directivity of the antenna.

2 Apparatus Required

Monopole and dipole antennas, RF signal generator, digital storage oscilloscope (DSO), spectrum analyzer, RF Tee connector, 50 Ω termination, antenna support stand, coaxial cables, and suitable RF adapters.

3 Theory

3.1 Basic Antenna Concepts

An antenna is a transducer that converts electrical signals into electromagnetic waves during transmission and converts incident electromagnetic waves into electrical signals during reception. The performance of an antenna can be characterized by its operating frequency, impedance matching, radiation pattern, polarization, beamwidth, and directivity.

A half-wave dipole consists of two conducting arms fed at the centre, with a total length approximately equal to half the operating wavelength. A quarter-wave monopole is approximately one-quarter wavelength long and is generally mounted above a conducting ground plane. The ground plane acts as an image plane, making the monopole behave approximately like half of a dipole antenna.

The wavelength of an electromagnetic wave is related to its frequency by

$$\lambda = \frac{c}{f}, \tag{1}$$

where λ is the wavelength, c is the speed of light in free space, and f is the frequency.

For an idealized quarter-wave monopole, the resonant length is approximately

$$L \approx \frac{\lambda}{4}. \tag{2}$$

Consequently, its theoretical resonant frequency is

$$f_{\text{th}} \approx \frac{c}{4L}, \tag{3}$$

where L is the physical length of the monopole. For a half-wave dipole, the corresponding approximation is $f_{\text{th}} \approx c/(2L)$, where L denotes the total dipole length.

These expressions provide approximate resonance frequencies. The actual resonant frequency can differ because of end effects, conductor diameter, feed arrangement, and the surrounding electromagnetic environment.

3.2 Antenna Radiation Pattern

The radiation pattern describes the variation of radiated power with direction. In a receiving experiment, the corresponding angular response can be investigated by measuring the received power while rotating the receiving antenna relative to the transmitting antenna.
A typical antenna radiation pattern consists of a main lobe, side lobes, and a back lobe. The main lobe corresponds to the direction of maximum radiation or maximum response, while side lobes and the back lobe represent responses in other directions.
The electric-field plane (E-plane) contains the electric-field vector and the direction of propagation. The magnetic-field plane (H-plane) contains the magnetic-field vector and the direction of propagation. These planes are perpendicular to each other.
The E-plane and H-plane patterns provide information about the directional characteristics of the antenna. Their angular widths can be used to estimate the directivity of the antenna.

3.3 Reflection Coefficient, VSWR, and Return Loss

When the antenna impedance differs from the characteristic impedance of the transmission line, a fraction of the incident electromagnetic signal is reflected back towards the source. This mismatch can be characterized by the voltage reflection coefficient:

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}, \tag{4}$$

where Z_L is the antenna input impedance and Z_0 is the characteristic impedance of the transmission line.
The magnitude of the reflection coefficient is denoted by $|\Gamma|$. For a perfectly matched antenna, $Z_L = Z_0$ and $|\Gamma| = 0$.
The **Voltage Standing-Wave Ratio** (VSWR) is defined as

$$\mathrm{VSWR} = \frac{1 + |\Gamma|}{1 - |\Gamma|}. \tag{5}$$

In the experimental arrangement, VSWR can be determined from the maximum and minimum voltage amplitudes observed during a frequency sweep:

$$\mathrm{VSWR} = \frac{V_{\max}}{V_{\min}}, \tag{6}$$

where $V_{\max}$ and $V_{\min}$ are the maximum and minimum voltage amplitudes, respectively.
The magnitude of the reflection coefficient can therefore be calculated using

$$|\Gamma| = \frac{\mathrm{VSWR} - 1}{\mathrm{VSWR} + 1} = \frac{V_{\max} - V_{\min}}{V_{\max} + V_{\min}}. \tag{7}$$

The return loss is defined as

$$\mathrm{RL} = -20 \log_{10} |\Gamma| \quad \mathrm{dB}. \tag{8}$$

A lower VSWR and a higher return-loss value indicate better impedance matching. In the frequency sweep, the frequency corresponding to the minimum VSWR is taken as an estimate of the antenna's operating or resonant frequency.

The frequency range over which

$$\text{VSWR} \leq 2 \tag{9}$$

is satisfied can be used as a practical measure of the antenna's impedance bandwidth.

3.4 Effect of Distance on Received Power

The received power depends on the separation between the transmitting and receiving antennas. Under free-space propagation conditions, the Friis transmission equation gives

$$P_r = P_t G_t G_r \left(\frac{\lambda}{4\pi R}\right)^2, \tag{10}$$

where P_t is the transmitted power, P_r is the received power, G_t and G_r are the gains of the transmitting and receiving antennas in the relevant directions, λ is the wavelength, and R is the separation between the antennas.
For fixed frequency, antenna gains, and transmitted power, the equation implies

$$P_r \propto \frac{1}{R^2}. \tag{11}$$

Thus, the received power decreases as the distance between the antennas increases. In ideal free space, doubling the distance reduces the received power to one-quarter of its original value.
Expressing the Friis equation in decibel units gives

$$\begin{aligned} P_r(\text{dBm}) = P_t(\text{dBm}) + G_t(\text{dBi}) + G_r(\text{dBi}) \\ + 20\log_{10}\left(\frac{\lambda}{4\pi R}\right). \end{aligned} \tag{12}$$

For fixed wavelength and antenna gains, the distance-dependent term is $-20\log_{10} R$. Therefore, a plot of received power in dB against $20\log_{10} R$ is expected to have a slope of approximately -1 under ideal free-space conditions.
In a laboratory environment, deviations from this relationship may occur because of reflections, interference, antenna misalignment, cable losses, and measurements made outside the far-field region.

3.5 Co-Polarization and Cross-Polarization

Polarization describes the orientation and evolution of the electric-field vector of an electromagnetic wave. For linearly polarized antennas, the relative orientation of the transmitting and receiving electric fields affects the received power.
In the co-polarization configuration, the transmitting and receiving antennas have aligned polarization directions. This configuration generally produces a stronger received signal when the antennas are also properly aligned in the direction of maximum response.
In the cross-polarization configuration, the polarization directions of the transmitting and receiving antennas are perpendicular. For ideal linearly polarized antennas, this arrangement produces zero received power because the electric field has no component along the receiving antenna's polarization direction.

For an ideal polarization mismatch angle ψ, the polarization coupling factor is

$$\eta_{\text{pol}} = \cos^2 \psi. \tag{13}$$

Hence, the received power due to polarization coupling can be expressed as

$$P_r(\psi) = P_{r,\max} \cos^2 \psi, \tag{14}$$

where $P_{r,\max}$ is the received power for aligned linear polarizations, assuming all other conditions remain unchanged.
For co-polarization, $\psi = 0^\circ$ and $\eta_{\text{pol}} = 1$. For ideal cross-polarization, $\psi = 90^\circ$ and $\eta_{\text{pol}} = 0$.
In practice, the cross-polarized signal may remain finite because of antenna imperfections, reflections, scattering, and limitations in the experimental alignment.

3.6 Half-Power Beamwidth

The half-power beamwidth (HPBW) is the angular width of the main lobe between the two directions at which the received power falls to half of its maximum value. These directions are known as the half-power or $-3\,\text{dB}$ points.
If $P_{\max}$ denotes the maximum received power, the half-power condition is

$$P(\theta) = \frac{P_{\max}}{2}. \tag{15}$$

In decibel units, this condition becomes

$$P_{\text{dB}}(\theta) = P_{\max,\text{dB}} - 3.01\,\text{dB}. \tag{16}$$

Let the two angular positions at which the half-power condition is satisfied be θ_1 and θ_2. The E-plane half-power beamwidth is

$$\theta_E = |\theta_2 - \theta_1|. \tag{17}$$

Similarly, the H-plane half-power beamwidth is

$$\theta_H = |\phi_2 - \phi_1|, \tag{18}$$

where ϕ_1 and ϕ_2 are the half-power angles in the H-plane.
A narrow HPBW indicates that the antenna's response is concentrated over a smaller angular region, whereas a broad HPBW indicates a wider angular response.

3.7 Antenna Directivity

Directivity quantifies the concentration of an antenna's radiation in a particular direction relative to an isotropic radiator. It depends on the angular distribution of radiated power and is distinct from antenna efficiency.
The directivity can be expressed in terms of the beam solid angle Ω_A as

$$D = \frac{4\pi}{\Omega_A}. \tag{19}$$

For an approximate estimate using the E-plane and H-plane half-power beamwidths, the beam solid angle can be approximated by their product:

$$\Omega_A \approx \theta_E \theta_H. \tag{20}$$

Consequently, the approximate directivity is

$$D \approx \frac{4\pi}{\theta_E \theta_H}, \tag{21}$$

where both θ_E and θ_H must be expressed in radians.
If the beamwidths are measured in degrees, they must first be converted using

$$\theta_{\text{rad}} = \theta_{\text{deg}} \frac{\pi}{180}. \tag{22}$$

The directivity calculated from Equation (21) is an approximate estimate. It assumes that the product of the two half-power beamwidths adequately represents the beam solid angle; the actual value depends on the complete three-dimensional radiation pattern.

4 Experimental Setup

The experimental arrangement consists of a transmitting antenna connected to an RF signal generator and a receiving antenna connected to a spectrum analyser. The transmitting and receiving antennas are mounted on suitable supports and aligned at the same height. Their relative orientation and separation are adjusted according to the measurement being performed.
The experiment is divided into three principal arrangements: frequency response and impedance matching, received power versus distance, and angular beam-pattern measurements to find the directivity of the antenna (DUT).

4.1 VSWR and Operating-Frequency Measurement

For the VSWR measurement, the RF signal generator, RF Tee connector, antenna under test (DUT), and digital storage oscilloscope are connected according to the laboratory setup.
The RF signal generator supplies a signal whose frequency is varied over the operating range of the antenna while its amplitude is kept constant. The RF Tee provides the measurement connection at the antenna feed. The oscilloscope is used to observe the voltage variation as the frequency is swept.
The maximum and minimum voltage amplitudes, $V_{\max}$ and $V_{\min}$, are recorded for each selected frequency. These measurements are used to calculate the VSWR, reflection coefficient, and return loss. The frequency corresponding to the minimum VSWR is taken as the approximate operating frequency of the antenna.
The oscilloscope input impedance is set to $1\,\mathrm{M}\Omega$ for the Tee-junction measurement, as specified in the laboratory manual.

4.2 Received-Power Measurement

For the distance-dependence experiment, the transmitting antenna is connected to the RF signal generator, and the receiving antenna is connected to the spectrum analyser.
A suitable operating frequency and signal amplitude are selected and kept constant during the measurement. The transmitting and receiving antennas are aligned with the same polarization and maintained at the same height.

4.3 Beam-Pattern Measurement

For the angular beam-pattern experiment, the transmitting and receiving antennas are separated by a distance of at least 3λ, as prescribed in the laboratory manual. The antennas are mounted at the same height and aligned for the selected polarization configuration.
The receiving antenna is rotated through the required angular range while the received power is measured using the spectrum analyser. The angular response is recorded separately in the E-plane and H-plane.

5 Experimental Procedure

5.1 Part A: VSWR and Antenna Operating Frequency

1. The RF signal generator, RF Tee connector, antenna under test (DUT), and digital storage oscilloscope were connected according to the VSWR measurement setup.

2. The oscilloscope input impedance was set to $1\,\mathrm{M}\Omega$.

3. The signal-generator amplitude was kept constant, and a suitable range of test frequencies was selected around the expected operating frequency of the antenna.

4. At each selected test frequency f_k, a small-span frequency sweep covering one complete amplitude-ripple period was performed.

5. During each sweep, the maximum and minimum voltage amplitudes, $V_{\max}$ and $V_{\min}$, respectively, were recorded. The same voltage measurement convention was used for both quantities.

6. The voltage standing-wave ratio was calculated using

$$\mathrm{VSWR} = \frac{V_{\max}}{V_{\min}}. \tag{23}$$

7. The magnitude of the reflection coefficient was calculated using

$$|\Gamma| = \frac{\mathrm{VSWR} - 1}{\mathrm{VSWR} + 1}. \tag{24}$$

8. The return loss was calculated in decibels:

$$\mathrm{RL} = -20 \log_{10} |\Gamma|. \tag{25}$$

9. The measurements were repeated at different frequencies across the expected operating range.

10. Graphs of VSWR versus frequency and return loss versus frequency were plotted. The frequency corresponding to the minimum VSWR was identified as the approximate operating frequency.

11. The frequency range over which $\mathrm{VSWR} \leq 2$ was determined.

5.2 Part B: Effect of Distance on Received Power

1. The transmitting and receiving antennas were connected to the RF signal generator and spectrum analyser, respectively.

2. A suitable frequency and signal amplitude within the operating range of the antenna were selected.

3. The monopole length was adjusted according to the selected operating frequency.

4. The transmitting and receiving antennas were aligned with the same polarization and maintained at the same height.

5. The initial separation between the antennas was set, and the received power was measured using the spectrum analyser.

6. The separation distance R was varied systematically. At each distance, the received power P_r was recorded while keeping the transmitting frequency and power fixed.

7. A graph of received power P_r versus distance R was plotted.

8. The quantity $20 \log_{10} R$ was calculated for each distance, and a second graph of received power in decibel units versus $20 \log_{10} R$ was plotted.

9. The variation in received power with increasing separation was examined. The reduction in received power when the distance was doubled was estimated from the observations.

5.3 Part C: Beam Pattern in Co-Polarization

1. The transmitting and receiving antennas were arranged at a separation of at least 3λ and maintained at the same height.

2. The antennas were aligned so that their electric-field directions were parallel, corresponding to the co-polarization configuration.

3. A fixed transmitting frequency within the operating range of the antenna was selected.

4. The receiving antenna was rotated through the E-plane angular range from -90° to $+90^\circ$. The received power was recorded at suitable angular intervals.

5. The receiving antenna was returned to its reference position.

6. The receiving antenna was then rotated through the H-plane angular range from -90° to $+90^\circ$, and the received power was recorded at the same suitable angular intervals.

7. Polar plots of the E-plane and H-plane beam patterns were constructed from the recorded angular measurements.

8. The measurements were repeated at two additional transmitting frequencies, and the resulting beam patterns were compared.

5.4 Part D: Beam Pattern in Cross-Polarization

1. The transmitting antenna was reoriented so that its electric-field direction was perpendicular to that of the receiving antenna.

2. The transmitting frequency, antenna separation, antenna heights, angular intervals, and instrument settings were kept the same as those used in the co-polarization measurements.

3. The receiving antenna was rotated through the E-plane angular range from -90° to $+90^\circ$, and the received power was recorded at each angular position.

4. The antenna was returned to its reference position and rotated through the H-plane angular range from -90° to $+90^\circ$. The received power was recorded at each position.

5. Polar plots of the E-plane and H-plane cross-polarization patterns were constructed.

6. The co-polarization and cross-polarization beam patterns were compared to examine the effect of polarization mismatch on the received power and angular response.

5.5 Part E: Half-Power Beamwidth and Directivity

1. The co-polarized E-plane beam pattern was examined, and the maximum received power in the main lobe was identified.

2. The two angular positions at which the received power fell to half its maximum value, corresponding to approximately $-3.01\,\mathrm{dB}$ relative to the peak, were determined.

3. The E-plane half-power beamwidth was calculated as the angular separation between these two positions:
$$\theta_E = |\theta_2 - \theta_1|. \tag{26}$$

4. The same procedure was applied to the co-polarized H-plane beam pattern to determine the H-plane half-power beamwidth:
$$\theta_H = |\phi_2 - \phi_1|. \tag{27}$$

5. Both beamwidths were converted from degrees to radians using
$$\theta_{\mathrm{rad}} = \theta_{\mathrm{deg}} \frac{\pi}{180}. \tag{28}$$

6. The approximate antenna directivity was calculated using
$$D \approx \frac{4\pi}{\theta_E \theta_H}, \tag{29}$$
where both beamwidths are expressed in radians.

7. The measured beamwidths and calculated directivity were tabulated and interpreted in terms of the directional characteristics of the antenna.

6 Observations

6.1 Observation Table: VSWR Measurement

Table 1: Frequency-sweep measurements for VSWR and return loss

Frequency f (MHz)	$V_{\max}$ (V)	$V_{\min}$ (V)	VSWR	Return loss (dB)
Yet to be calculated.				

6.2 Recorded Antenna-Length Calculation

The notebook gives the monopole length as

$$L = 5.6\,\text{cm} = 0.056\,\text{m}. \tag{30}$$

For an ideal quarter-wave monopole,

$$L = \frac{\lambda}{4}. \tag{31}$$

Therefore,

$$\begin{aligned} \lambda &= 4L && (32)\\ &= 4(5.6\,\text{cm}) && (33)\\ &= 22.4\,\text{cm} && (34)\\ &= 0.224\,\text{m}. && (35) \end{aligned}$$

The theoretical resonance frequency is

$$\begin{aligned} f_{\text{th}} &= \frac{c}{\lambda} && (36)\\ &= \frac{3.0 \times 10^8}{0.224} && (37)\\ &= 1.3393 \times 10^9\,\text{Hz}. && (38) \end{aligned}$$

Hence,

$$\boxed{f_{\text{th}} \approx 1.3393\,\text{GHz}} \tag{39}$$

The theoretical value of operating frequency is:

$$\boxed{f_{\text{exp}} \approx 1.42\,\text{GHz}}. \tag{40}$$

6.3 Observation Table: Received Power versus Distance

For each recorded distance, calculate

$$X_R = 20 \log_{10} R, \tag{41}$$

where R is expressed numerically in centimetres for consistency with the recorded distance table. The logarithmic distance is used for the second graph, with received power plotted in dBm against $20\log_{10} R$.
The distance values above are transcribed from the notebook. The readings at 422, 472, and 522 cm do not follow a strictly decreasing trend in received power. They have been retained as recorded rather than altered to enforce the theoretical inverse-square relationship.

Table 2: Received power at different antenna separation distances

S. No.	Distance R (cm)	$X_R = 20log_{10}R$	Received power P_r (dBm)
1	72	37.15	−37.81
2	122	41.73	−43.25
3	172	44.71	−46.50
4	222	46.93	−48.72
5	272	48.69	−48.81
6	322	50.16	−50.89
7	372	51.41	−56.33
8	422	52.51	−51.67
9	472	53.48	−59.00
10	522	54.35	−53.24
11	574	55.18	−58.00

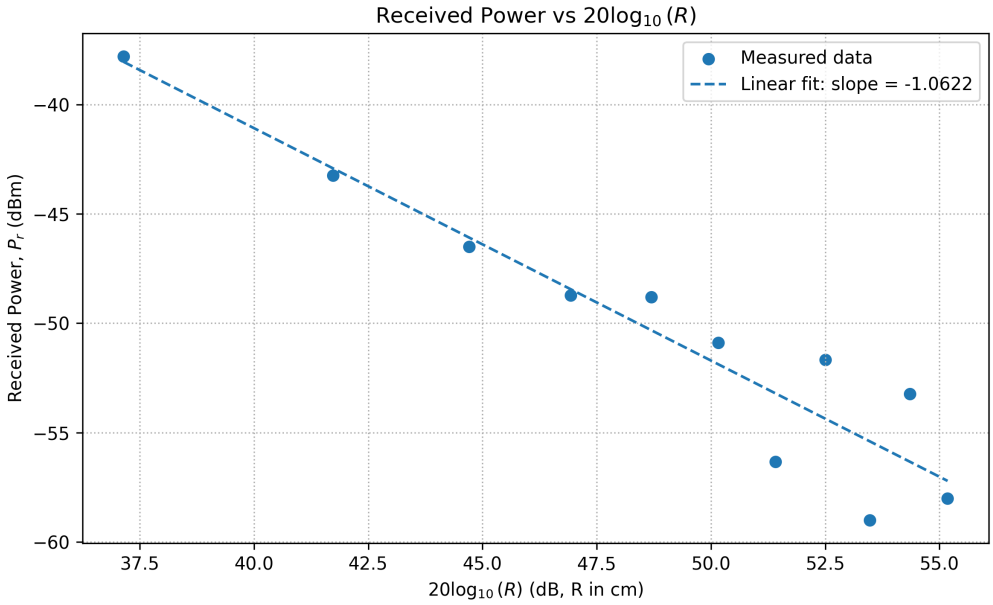


Figure 1: Received Power as a function of $20log_{10}R$

6.4 Observation Table: Co-Polarization

Table 3: Co-polarization measurements at different angular positions.

Angle (°)	P_H (dBm)	P_E (dBm)
0	-41.11	-61.64
10	-41.61	-67.41
20	-43.33	-64.69
30	-45.70	-63.38
40	-48.58	-63.32
50	-49.80	-61.873
60	-53.30	-61.20
70	-58.04	-62.26
80	-58.77	-65.47
90	-69.10	-71.24
100	-84.30	-76.54
110	≈ -74.05	-74.29
120	-83.50	-67.50
130	-73.70	-64.41
140	-62.00	-66.20
150	-61.90	-68.12
160	-61.96	-69.27
170	-60.09	-73.75
180	-62.00	-71.05

Angle (°)	P_H (dBm)	P_E (dBm)
190	-64.50	-64.57
200	-60.40	-67.79
210	-58.80	-68.14
220	-58.95	-66.84
230	-58.49	-74.56
240	-60.88	-72.30
250	-62.50	-67.81
260	-64.45	-63.07
270	-74.52	-66.91
280	-74.15	-69.67
290	-61.78	-63.13
300	-58.41	-61.79
310	-55.90	-60.97
320	-52.47	-58.53
330	-49.15	≈ -57.22
340	-46.10	-56.77
350	-43.43	-58.45
360	-42.53	-58.33

6.5 Observation Table: Cross-Polarization

Table 4: Cross-polarization measurements at different angular positions.

Angle (°)	P_H (dBm)	P_E (dBm)
0	-50.55	-44.78
15	-51.83	-44.86
30	-55.27	-46.71
45	≈ -61.32	-50.08
60	-71.60	-53.90
75	-70.42	-58.10
90	-62.86	-61.23
105	-66.11	-57.72
120	-70.97	-62.12
135	-66.54	≈ -66.59
150	-76.50	-66.15
165	-71.56	-62.70
180	-69.49	-63.27

Angle (°)	P_H (dBm)	P_E (dBm)
195	-68.05	-65.89
210	≈ -63.72	-65.90
225	-73.74	-65.23
240	-77.35	-63.60
255	-71.47	-61.50
270	-63.14	-59.80
285	-58.35	-61.40
300	-58.70	-54.22
315	-56.35	-51.28
330	-55.02	≈ -48.18
345	-52.97	-45.85
360	-50.73	-44.33

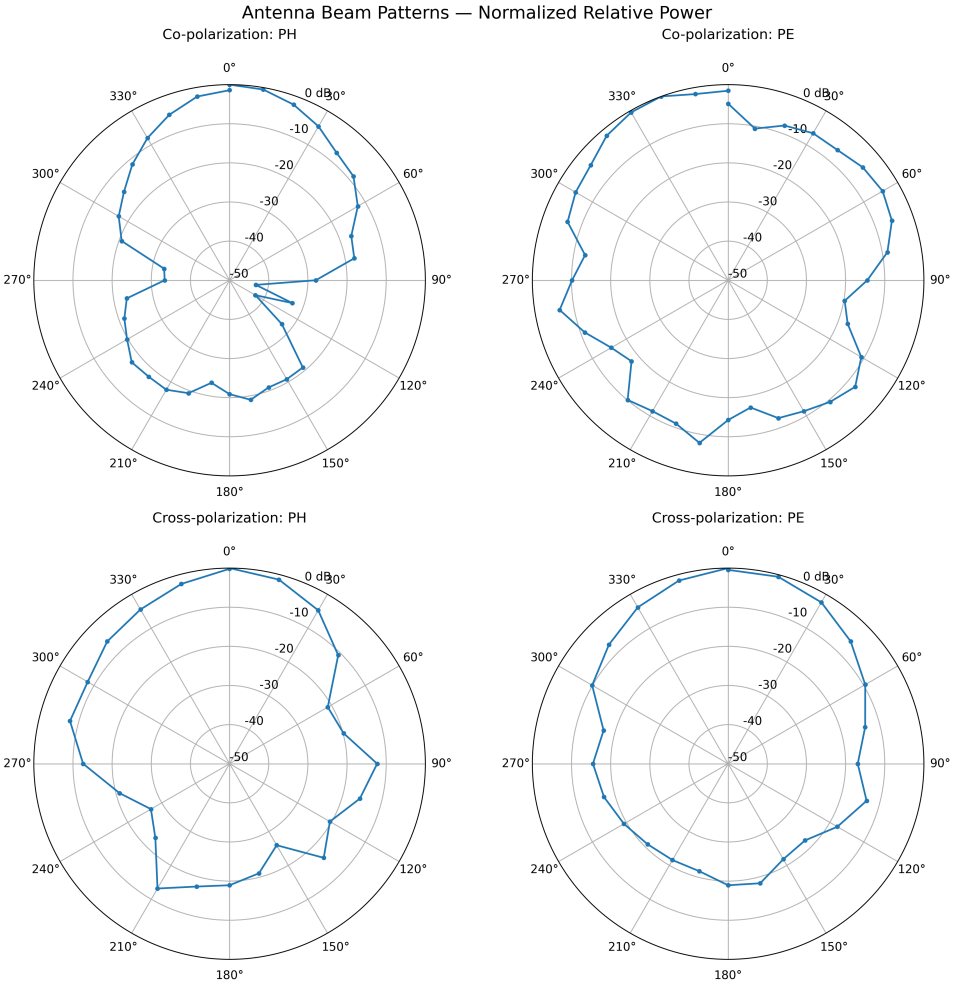


Figure 2: Beam Patterns for Co and Cross Polarization of Horn Antenna (DUT)

Table 5: Summary of recorded experimental observations

Quantity	Recorded information
Monopole length	5.6 cm
Theoretical wavelength	22.4 cm
Theoretical resonance frequency	Approximately 1.3393 GHz
Experimentally noted operating frequency	Approximately 1.42 GHz
Distance-measurement interval	72–574 cm
Number of distance readings	11
Co-polarization angular interval	10°
Cross-polarization angular interval	15°
Co-polarization angular range	0°–360°
Cross-polarization angular range	0°–360°
VSWR frequency-sweep readings	Yet to be calculated

7 Results

7.1 Antenna Operating Frequency

The recorded monopole length was 5.6 cm. Assuming an ideal quarter-wave monopole, the theoretical resonance frequency was estimated using

$$f_{\text{th}} = \frac{c}{4L}. \tag{42}$$

Substituting $L = 0.056\,\text{m}$ gives

$$f_{\text{th}} = \frac{3.0 \times 10^8}{4(0.056)} \approx 1.3393\,\text{GHz}. \tag{43}$$

The operating frequency recorded in the observation notebook was approximately 1.42 GHz. The difference between the recorded experimental estimate and the theoretical value is

$$\Delta f = |f_{\text{exp}} - f_{\text{th}}| \approx 80.7\,\text{MHz}. \tag{44}$$

The percentage difference relative to the theoretical value is

$$\%\text{ difference} = \frac{|f_{\text{exp}} - f_{\text{th}}|}{f_{\text{th}}} \times 100 \approx 6.02\%. \tag{45}$$

The difference may be associated with the approximate quarter-wave model, antenna end effects, feed configuration, and the surrounding experimental environment. Since the numerical frequency-sweep readings are unavailable, the minimum VSWR and the frequency range for which VSWR ≤ 2 could not be independently determined.

7.2 Effect of Distance on Received Power

The received-power measurements were recorded for antenna separations ranging from 72 cm to 574 cm. The recorded received power varied from -37.81 dBm to -59.00 dBm. The largest recorded received power occurred at 72 cm, while the smallest occurred at 472 cm. Overall, the received power tended to decrease as the separation increased, although the readings were not strictly monotonic.

According to the Friis transmission equation, for fixed frequency, antenna gains, and transmitted power,

$$P_r \propto \frac{1}{R^2}. \tag{46}$$

Consequently, the ideal free-space received power in decibel units should vary approximately linearly with $20 \log_{10} R$, with a slope of -1 when received power is plotted in dBm against $20 \log_{10} R$.

The non-monotonic measurements indicate deviations from a simple inverse-square trend. Possible contributors include multipath interference, reflections from nearby objects, antenna alignment, and measurement uncertainty.

7.3 Co-Polarization Beam Pattern

The co-polarization observations were recorded at 10° angular intervals over the range 0° to 360°. The notebook contains two measured power columns, P_H and P_E, both expressed in dBm.

The recorded values show angular variations in received power. The P_H measurements range from approximately $-41.11\,\text{dBm}$ to $-84.30\,\text{dBm}$, while the P_E measurements range from approximately $-56.77\,\text{dBm}$ to $-76.54\,\text{dBm}$.

These variations demonstrate that the received signal depends on antenna orientation. However, the exact E-plane and H-plane beamwidths cannot be established reliably until the measurement columns and their corresponding angular configurations are confirmed.

7.4 Cross-Polarization Beam Pattern

The cross-polarization observations were recorded at 15° angular intervals over the range 0° to 360°. The two recorded power columns, P_H and P_E, exhibit variations with angular position.

The recorded P_H values range approximately from $-50.55\,\text{dBm}$ to $-77.35\,\text{dBm}$. The recorded P_E values range approximately from $-44.33\,\text{dBm}$ to $-66.59\,\text{dBm}$.

The measurements show that the received signal remains finite in the cross-polarization configuration. This can arise from imperfect polarization alignment, antenna imperfections, and reflections in the experimental environment.

A quantitative cross-polarization discrimination cannot be determined reliably from these ranges alone because corresponding co-polarization and cross-polarization measurements must be compared at the same angles and under identical conditions.

8 Key Takeaways

1. **Quarter-wave resonance:** For the recorded monopole length of 5.6 cm, the ideal quarter-wave approximation gives a theoretical resonance frequency of approximately 1.3393 GHz. The actual operating frequency is 1.42 GHz, (which is the frequency for 21 *cm* HI line), giving an error of about 6.02%.

2. **Impedance matching:** VSWR and return loss are useful indicators of the impedance match between an antenna and its transmission line. A reliable experimental evaluation requires the measured maximum and minimum voltage amplitudes over the frequency sweep.

3. **Distance dependence:** The received power generally decreased as the separation between the transmitting and receiving antennas increased. The measurements showed fluctuations around the expected free-space inverse-square dependence.

4. **Angular response:** Both sets of angular observations exhibited variations in received power with antenna orientation. These measurements can be used to construct beam patterns after the recorded power columns and measurement planes are verified.

5. **Polarization:** The co-polarization and cross-polarization configurations illustrate the importance of electric-field orientation in antenna reception. Quantitative comparison requires measurements at corresponding angles under consistent experimental conditions.

6. **Beamwidth and directivity:** The half-power beamwidths describe the angular extent of the main lobe. The E-plane and H-plane beamwidths can be combined to obtain an approximate directivity, provided the correct half-power points are identified from the measured patterns.

7. **Experimental uncertainty:** Reflections, multipath propagation, alignment errors, finite angular resolution, and instrument uncertainty can influence the observed results. Such effects should be considered when comparing experimental measurements with ideal antenna theory.